ISOTOPIA-2.23

Simulation of medical isotope production

Arjan Koning

s

nds.iaea.org/talys

Typesetting: The Legrand Orange Book, LaTeX Template, Version 2.1.1 (14/2/16), downloaded from: `www.LaTeXTemplates.com`. Original author: Mathias Legrand (legrand.mathias@gmail.com) with modifications by: Vel (vel@latextemplates.com). License: CC BY-NC-SA 3.0 `creativecommons.org/licenses/by-nc-sa/3.0/`.

March 2026

About the author

Arjan Koning is a nuclear physicist with a Masters Degree in Theoretical physics at the Univ. of Amsterdam, a PhD in the Natural Sciences on Multi-step direct reactions at the Univ. of Groningen, and a Professorship at the Univ. of Uppsala on theoretical nuclear reactions.
Arjan is currently Head of the Nuclear Data Section at the IAEA in Vienna. Before that, he has worked at ECN/NRG Petten, the Netherlands, on nuclear reaction data for science and technology, and as guest scientist at CEA/Bruyères-le-Châtel and Los Alamos National Laboratory on the development and computational implementation of nuclear reaction models. He has led several students to PhD degrees, has coordinated and chaired various international nuclear data projects such as the OECD/NEA JEFF and WPEC projects, and has advised governments and international organisations on nuclear research and development. Among his scientific accomplishments are innovations in nuclear reaction physics, especially for the optical model and pre-equilibrium reactions, and initiating the '3 T's': the TALYS nuclear model code, Total Monte Carlo uncertainty propagation and the TENDL nuclear data library. He has been awarded the Oppenheimer Coin from Los Alamos National Laboratory, and the title of Doctor Honoris Causa by Uppsala University. As of 2025, his h-index is 62 with more than 26 000 citations.
Although Arjan is currently in a managerial role, he aims to keep his scientific creativity alive by maintaining and extending TALYS plus all software that emerges from that. Pleas from his friends to also spend time on other things are sometimes honoured.

Preface

ISOTOPIA is a software package for the prediction of medical isotope production. In principle, it can calculate isotope yields for any production route based on incident neutrons, photons, protons, deuterons, tritons, helions or alpha particles, and for either natural or enriched targets. Most of the computational ingredients needed for such a simulation are straightforward. The decisive ingredients, however, are reliable nuclear reaction cross sections and radioactive decay data. In ISOTOPIA, the IAEA medical isotope data library for about 150 reaction channels is combined with TENDL-2025 and, for photon-induced reactions, TENDL-2023, so that both well-established production routes and less explored possibilities can be treated in one calculational scheme.

The motivation for ISOTOPIA is the expectation that medical radioisotopes produced with both reactors and accelerators will remain necessary in the foreseeable future, if medical diagnosis and therapy are to remain both successful and economically affordable. This issue has become more urgent in recent years, in particular because several older isotope-production reactors need to be replaced. Accelerators, traditionally used mainly for neutron-poor PET isotopes and for several therapeutic isotopes, are now also discussed as possible production devices for "reactor nuclides", with the ^{99}Mo/^{99m}Tc generator as the most prominent example. The choice between such production routes has economical, logistic and political aspects, but the present code is meant to address the corresponding nuclear-science question: what is produced, with which yield, and with which impurities, for a specified irradiation route? The reliability of this answer is inevitably governed by the quality of the nuclear reaction cross sections.

At certain moments in time, a well-defined version of ISOTOPIA is frozen. You are now reading the tutorial of version 2.23. Future versions should become more reliable through improvements of the ISOTOPIA code itself and, equally important, through improved cross section and radioactive decay data libraries.

ISOTOPIA is the calculation engine behind the Medical Isotope Browser, nds.iaea.org/mib, which provides a graphical interface for the efficient analysis of production routes.

License, contact and reference

As mentioned on the first page and in the source code, ISOTOPIA falls in the category of MIT License software.

In addition to the MIT *terms* I have a *request*:

- When ISOTOPIA is used for your reports, publications, etc., please make a proper reference to the code. At the moment this is:
  A.J. Koning, ISOTOPIA: Simulation of medical isotope production with accelerators, unpublished tutorial (2025).

The webpage for ISOTOPIA is **nds.iaea.org/talys**.

Acknowledgements

I wish to thank a few persons who have contributed to the present state of ISOTOPIA:

- Marieke Duijvestijn, who set up the first calculation routes with the TALYS code for isotopic yields, which we needed for a project on the production of ^{68}Ga, when we were still working at NRG Petten,
- Marco Verpelli, for integrating ISOTOPIA in the Medical Isotope Browser and for suggesting the associated improvements to the code,
- Steven van der Marck, for his initial help in constructing a software tool like this and for useful discussions,
- Will Webster and Bernard Erasmus, for important feedback on the implementation of ISOTOPIA in the TALYS code,
- Roberto Capote Noy and the entire consortium responsible for the IAEA medical isotope data library, for helpful discussions and for providing the specific data library for about 150 isotopes,
- Natalie Gaughan, for testing many of the isotope yield calculations with the TALYS code,
- Ulli Koester, for taking a critical look at our Medical Isotope Browser and suggesting improvements.

Arjan Koning

Contents

1. Introduction

The current worldwide market for medical isotopes is estimated to be of the order of 6–8 billion dollars per year and is expected to increase to about 15 billion dollars in 2035. This development places high demands on isotope production capacity, not only in terms of the number of facilities, but also in terms of the diversity of production routes. Accelerators, research reactors and other neutron sources all contribute to this supply chain, and each of them requires a quantitative understanding of the nuclear reactions by which the isotopes are produced.

The essential feature of a medical isotope is that it releases radiation in the form of alpha, beta or gamma particles, which can be used for diagnosis, therapy or a combination of both, i.e. theranostics. The isotope therefore has to be radioactive and, in most practical cases, it has to be formed by nuclear transmutation. There are three main irradiation sources for medical isotope production:

- charged-particle accelerators, with projectile types ranging from protons, which dominate present use, to heavier ions,
- photonuclear reactions,
- neutron sources, usually research reactors.

The nuclear reactions induced by these sources can broadly be divided into two classes:

- activation reactions, for which the produced isotope is relatively close to the target and is formed through a comparatively simple reaction path,
- fission reactions leading to fission products, with the production of ^{99}Mo in research reactors as the best known example.

For diagnosis, almost all radioisotopes belong to the categories of single photon emission computed tomography (SPECT) or positron emission tomography (PET). Although PET scans have important qualitative advantages and their market share is increasing, the ease of use and cost effectiveness of SPECT, with the ^{99}Mo/^{99m}Tc generator as its workhorse, imply that its dominant role is unlikely to disappear in the coming decade. As a useful rule of thumb, while realizing that exceptions exist, neutron-rich nuclides obtained as fission products or by neutron capture in a reactor are often used for SPECT, whereas neutron-poor nuclides produced with accelerators are used for PET. For therapy, both reactor-produced and accelerator-produced isotopes are relevant.

The current version of ISOTOPIA is devoted to isotope production with charged particles, photons and neutrons. A typical use case starts with a radioisotope that has already been identified as promising for medical diagnosis or therapy on the basis of its biological, chemical and decay properties. The subsequent question, "What is the best way to produce this isotope?", does not usually have a single yes-or-no answer. It decomposes into several more specific questions, such as:

- What is the optimal production route, i.e. what is the best target-projectile combination?
- For charged-particle irradiation, what are the optimal incident beam energy and target thickness, or equivalently, what is the projectile energy interval inside the target?
- What are the optimal irradiation time and cooling period?
- What is the isotopic yield during irradiation, directly after irradiation, and after a specified cooling period?
- What are the elemental and isotopic contaminations after irradiation and after cooling?
- How strongly do these answers depend on using an enriched rather than a natural target?

To answer such questions, analytical models have been implemented which parameterize the main characteristics of a reactor or accelerator source and its target. These models calculate isotopic yields for different production routes as a function of irradiation and cooling time. Such a systematic treatment has become practical because modern cross section and decay data libraries now cover a very large part of the relevant projectile-target-product space.

The nuclear-physics aspects of these questions are not the entire story. Several economical and logistic questions, often strongly correlated, also have to be answered, for example:

- How much irradiation is needed to meet the present or expected demand?
- What are the logistical implications of the half-life of the desired product and of its purity, which may depend strongly on the production route?
- What are the costs of producing a sufficiently enriched target?

In the end, the decision to build or use a production facility may also depend on political considerations. These aspects are not treated here. The purpose of this tutorial is to describe the scientific and computational part of the problem.

As specific features of ISOTOPIA we mention

- Estimation of isotope yields for a user-specified irradiation source and energy range, target thickness, irradiation time and cooling period. Incident neutrons, photons, protons, deuterons, tritons, helions and alpha particles are covered for arbitrary target materials.
- Activities of all residual products as a function of irradiation time. When appropriate, the results are averaged over time to obtain average yields in [MBq/mAh] or [MBq/(g.h)].
- Estimation of the simultaneous production of radioactive contaminations.
- Robustness: by combining the complete TENDL data library with well-evaluated reaction channels from the IAEA medical isotope library, ISOTOPIA can be used for essentially any isotope production route of interest. It is therefore a general predictive tool, with a basic quality level tested over the full range of implemented cases.
- A transparent source program.
- Input and output communication which is intended to be easy to use and understand.
- A user tutorial.
- A set of sample cases.

The starting point of all calculations is the set of cross sections for the relevant reaction channels. These cross sections are obtained from two sources:

- the TENDL-2025 nuclear data library [1], and TENDL-2023 for photons, produced by the nuclear reaction model code TALYS [2] and associated nuclear data tools,
- the IAEA medical isotope database, which contains high-quality in-depth evaluations of

important reaction channels based on available experimental data [3, 4, 5, 6].

The present tutorial describes the ISOTOPIA code and its input and output. We stress, however, that a graphical user interface has been built around the code at the IAEA: the Medical Isotope Browser, nds.iaea.org/mib. This browser drives ISOTOPIA and is often the most efficient way to study medical isotope production as a function of irradiation characteristics.

1.1 How to use this tutorial

Although we would be honored if the tutorial were read from beginning to end, not all parts will be necessary, relevant or suitable for every user. The contents are as follows:

Chapter 2: Installation guide for ISOTOPIA.
Chapter 3: Main input rules.
Chapter 4: General formalism for isotope production.
Chapter 5: Nuclear reaction and structure information needed in the form of cross section and decay data.
Chapter 6: Reference guide with all keywords.
Chapter 7: Sample cases and verification.
Chapter 8: Outlook and conclusions.

2. Installation and getting started

2.1 The ISOTOPIA package

In what follows we assume ISOTOPIA will be installed on a Linux or MacOS operating system. The total ISOTOPIA package is in the *isotopia/* directory and contains the following directories and files:

- *LICENSE* is the license file,
- *README.md* outlines the contents of the package.
- *install_isotopia.bash*, *code_build.bash* and *path_change.bash* are scripts that take care of the installation,
- *source/* contains the Fortran source code of ISOTOPIA
- *files/* contains the natural abundances and the decay data library.
- *doc/* contains the documentation: this tutorial in pdf format.
- *samples/* contains the input and output files of the sample cases.

Outside *isotopia/*, you need to have the directory *isotopia.libs/* which contains all cross section data. In fact the place of this directory can be set in the *machine.f90* subroutine.

In total, you will need about 20 Gb of free disk space to install ISOTOPIA, which is almost entirely due to the cross section database in the *isotopia.libs/* directory. The code has so far *only* been tested on various Linux and MacOS systems, so we can not guarantee that it works on all operating systems. The only OS dependencies we can think of are the directory separators '/' we use in pathnames that are hardwired in the code. If there is any dependence on the operating system, the associated statements can be altered in the subroutine *machine.f90*.

ISOTOPIA has been successfully tested for the gfortran compiler.

2.2 Installation

The installation of ISOTOPIA is straightforward. You can download ISOTOPIA via either git

- **git clone https://github.com/arjankoning1/isotopia.git**

or by getting the tar file
- from **https://nds.iaea.org/talys/isotopia.tar**
- **tar zxf isotopia.tar**

We here provide the necessary steps to do the installation, For a Unix/Linux system, the installation is expected to be done by the simple command
- *install_isotopia.bash*

which calls the scripts
- *code_build.bash* where you may set, if needed, the name of your compiler and the place where you want to store the ISOTOPIA executable.
- **path_change.bash** to set the pathname in the source code.

An alternative installation option is
- **cd isotopia/source**
- **make**

If this does not work for some reason, we here provide the necessary steps to do the installation manually. For a Unix/Linux system, the following steps should be taken:
- **cd isotopia/source** If *code_build.bash* has not already replaced the path name in *machine.f90*, do it yourself. We think this is the only Unix/Linux machine dependence of ISOTOPIA. We expect no complaints from the compiler.
- **gfortran -c *.f90**
- **gfortran *.o -o isotopia**
- **mv isotopia ../bin**

The above commands represent the standard compilation options. Consult the tutorial of your compiler to get an enhanced performance with optimization flags enabled. We note that an ISOTOPIA run generally takes only less than a second so extreme optimization is not necessary.

2.3 Verification

If ISOTOPIA is installed, testing the sample cases is the logical next step. The *samples/* directory contains the script *verify* that runs all the test cases. Each sample case has its own subdirectory, which contains a subdirectory *org/*, where we stored the input files and *our* calculated results. It also contains a subdirectory *new*, where we have stored the input files only and where the *verify* script will produce *your* output files. A full description of the keywords used in the input files is given in Chapter 3. Chapter 7 describes all sample cases in full detail. Note that in each subdirectory a file with differences with our original output is created.

2.4 Getting started

If you have created your own working directory with an input file named e.g. *isotopia.inp*, then a ISOTOPIA calculation can easily be started with:

isotopia $<$ **isotopia.inp** $>$ **isotopia.out**

where the names *isotopia.inp(out)* are not obligatory: you can use any name for these files. Besides the general output file *isotopia.out* various other output files, with activities per isotope,are produced.

3. Input description

For the communication between ISOTOPIA and its users, we have constructed an input/output method which shields beginners from all the possible options for parameters that can be specified in ISOTOPIA, while enabling at the same time maximal flexibility for experienced users.

An input file of ISOTOPIA consists of keywords and their associated values. Before we list all the input possibilities, let us illustrate the use of the input by the following example. It represents a minimum input file for charged particle irradiation:

```
projectile p
element    Mo
mass       100
ebeam     16.
```

This input file represents the simplest question that can be asked to ISOTOPIA: if a 100 % enriched ^{100}Mo target is bombarded by 16 MeV protons, what isotopes are produced as a function of time, and what is the radioactive yield? Behind this simple input file, however, there are several default values for the various assumptions, parameters, output flags, etc., that you may or may not be interested in. When you use a minimal input file like the one above, you leave it to the author of ISOTOPIA to choose all the parameters for you, including accelerator and target specification, as well as the level of detail of the output file. If you want to use specific parameters other than the default, or want to have more specific information in the output file(s), more keywords are required. Obviously, more keywords means more flexibility and, in the case of adequate use, better results. For example, one probably would like to enter the accelerator beam current, target size and material density in the input. In this Chapter, we will first give the basic rules that must be obeyed when constructing an input file for ISOTOPIA.

3.1 Basic input rules

Theoretically, it would be possible to make the use of ISOTOPIA completely idiot-proof, i.e. to prevent the user from any input mistakes that possibly can be made and to continue a calculation with "assumed" values. Although we have invested a relatively large effort in the user-friendliness of ISOTOPIA, we have not taken such safety measures to the extreme limit and ask at least some minimal responsibility from the user. Once you have accepted that, only very little effort is required to work with the code. Successful execution of ISOTOPIA can be expected if you stick to the following simple rules and possibilities of the input file:

1. One input line contains one keyword. Usually it is accompanied by only one value, but some keywords (e.g. **Tcool**) need to be accompanied by more than one value on the same line.
2. A keyword and its value(s) *must* be separated by at least 1 blank character.
3. The keywords can be given in arbitrary order. If you use the same keyword more than once, the value of the last one will be adopted.
4. All characters can be given in either lowercase or uppercase.
5. A keyword *must* be accompanied by a value. To use default values, the keywords should simply be left out of the input file.
6. An input line starting with a # in column 1 is neglected. This is helpful for including comments in the input file or to temporarily deactivate keywords.
7. A minimal input file always consists of 3 or 4 lines and contains the keywords **projectile**, **element**, **mass** while for charged particle irradiation also **ebeam** needs to be given. These keywords *must* be given in any input file.
8. An input line may not exceed 132 characters.

As an example of rules 3, 4, and 6, it can be seen that the following input file is completely equivalent to the one given in the beginning of this chapter:

```
Ebeam     16.
Element    Mo
MASS       100
projectile p
#rho 5.
```

In the following erroneous input file, only the second and third lines are correct, while rules 2, and 5 are violated in the other lines.

```
Ebeam22.
projectile p
MASS 103
Element
```

In cases like this, ISOTOPIA will give a specific error message for the first encountered problem and the execution will be stopped. We like to believe that we have covered all such cases and that it is impossible to let ISOTOPIA crash (at least with our compilers) without giving an appropriate error message, but you are of course invited to prove and let us know about the contrary (Sorry, no cash rewards). Typing errors in the input file should be spotted by ISOTOPIA, e.g. if you write **projjectile d**, it will tell you the keyword is not in our list, and the code will gracefully stop.

4. Radioisotope production

This chapter describes the formalism used to calculate the production yield of a radioactive nucleus by a nuclear reaction. Starting from the general equations, detailed in the Appendix, one can introduce realistic approximations and derive simpler expressions which are sufficient for almost all activation cases of interest. In ISOTOPIA we have implemented analytical solutions of the production and depletion equations. These are expected to be adequate for nuclides produced by activation.

4.1 Equations for production and depletion of nuclides

The production and depletion of radionuclides can be described by a system of Bateman equations for the number of nuclides N_i of type i at time t,

$$\frac{dN_i(t)}{dt} = \sum_k (\lambda_{ki} + R_{ki}) N_k(t) - (\lambda_i + \sum_j R_{ij}) N_i(t), \tag{4.1}$$

where N_i can be produced by nuclear reactions and decay from all feeding nuclides k, with reaction constants R and decay constants λ, and is simultaneously depleted through reactions and decay to all nuclides j. The most general solution of this system is obtained with matrix exponential solvers. However, for almost all activation cases relevant for radioisotope production, and as shown in the Appendix, well-justified assumptions lead to the following time dependence during irradiation:

$$\begin{aligned}
N_T(t) &= N_0 \, e^{-R_T t}, \\
N_p(t) &= N_0 \, \frac{R_{Tp}}{\lambda_p - R_T} \, [e^{-R_T t} - e^{-\lambda_p t}], \\
N_i(t) &= N_0 \, \frac{R_{Ti}}{\lambda_i - R_T} \, [e^{-R_T t} - e^{-\lambda_i t}] + N_0 \frac{R_{Tp}\lambda_{pi}}{\lambda_p - R_T} \left[\frac{e^{-R_T t} - e^{-\lambda_i t}}{\lambda_i - R_T} - \frac{e^{-\lambda_p t} - e^{-\lambda_i t}}{\lambda_i - \lambda_p} \right],
\end{aligned} \tag{4.2}$$

where

- N_T is the number of target nuclides,
- N_i is the number of produced nuclides of interest,
- N_p is the number of parent nuclides which decay to nuclide i,
- $N_0 = N_T(t=0)$ is the initial number of target nuclides,
- λ_i is the total decay constant per atom for nuclide i, with $\lambda_i = \ln 2/T_{i,1/2}$ and $T_{i,1/2}$ the half-life. The decay constant can be interpreted as the probability per unit time for a single atom to decay and has dimension s^{-1},
- λ_{pi} is the decay constant per atom for parent nuclide p decaying to nuclide i,
- R_T is the total reaction constant per atom for target nuclide T, which can be interpreted as the probability per unit time that a single target atom undergoes a nuclear reaction and has dimension s^{-1},
- R_{Ti} is the reaction constant per atom for the reaction from target nuclide T to nuclide i.

In these equations, the total decay constant of nuclide i is the sum over the partial decay constants to all K daughter nuclides,

$$\lambda_i = \sum_{k=1,k\neq i}^{K} \lambda_{ik}. \tag{4.3}$$

In most applications, the loss term feeds only one or two channels, namely beta decay to the ground state or an isomer of the daughter nuclide d,

$$\lambda_i = \lambda_{id} = \lambda_{id}^{g.s.} + \lambda_{id}^{iso}. \tag{4.4}$$

The total nuclear reaction constant per atom for nuclide T is

$$R_T = \sum_{k=1,k\neq T}^{K} R_{Tk}. \tag{4.5}$$

Equation (4.2), derived in the Appendix, is more general than what is needed in most activation analyses of radioisotope production. In many practical cases, the relevant term is simply that of a radioactive nuclide produced directly from the target and decaying afterwards,

$$N_i(t) = N_0 \, \frac{R_{Ti}}{\lambda_i - R_T} \, [e^{-R_T t} - e^{-\lambda_i t}]. \tag{4.6}$$

The reaction constant R_T is often very small on the time scale of the irradiation, typically of the order of 10^{-10} s^{-1}, so that $N_T(t) \simeq N_0$. In that limit, target depletion is commonly neglected, although it should be retained for very high burn-up. With negligible R_T, Eq. (4.6) reduces further to

$$N_i(t) = N_0 \, \frac{R_{Ti}}{\lambda_i} [1 - e^{-\lambda_i t}], \tag{4.7}$$

which is the expression used in many simple activation analyses.

In ISOTOPIA we have implemented the full set of Eq. (4.2). The first line accounts for burn-up of the target, while the second and third lines include the possible production of the nuclide of interest through one indirect decay route, called here the parent route.

If formation through a parent isotope is neglected, the irradiation time for which a maximum yield is obtained can be derived by setting the derivative of $N_i(t)$ in Eq. (4.6) to zero. This gives

$$t_{max} = \frac{ln(\lambda_i/R_T)}{\lambda_i - R_T}. \tag{4.8}$$

The number of target atoms available for activation at $t = 0$, N_0, for $1 \leq m \leq M$, where M is the number of natural isotopes constituting the target element, is

$$N_0 = \frac{N_A}{A} B_m M_{tar}, \tag{4.9}$$

where $N_A = 6.022\ 10^{23}$ is Avogadro's number, A is the mass number, B_m is the abundance of isotope m, with $\sum_m^M B_m = 1$, and M_{tar} is the target mass in [g]. This mass is obtained from the mass density ρ in [g/cm^3] and the target volume V_{tar} in [cm^3],

$$M_{tar} = \rho\ V_{tar}. \tag{4.10}$$

The formalism given above is still general with respect to the irradiation source and applies to neutron, charged-particle and photon irradiation. The quantity that differs among these cases is the reaction constant R. It depends on two essential ingredients: the geometry-dependent aspects of the irradiation and the cross sections. Nuclear data are therefore central to the formalism: decay data determine λ, while reaction cross sections determine R. These data are discussed in the next chapter.

4.2 Reaction constants and rates

For an external spectrum, such as that of a photon or neutron source, the reaction constant per atom for producing isotope i is the product of the total particle flux, ϕ^{tot} in [cm^{-2}s^{-1}], and the average of the residual-production cross section σ_i over the normalized energy-dependent source spectrum f,

$$R_{Ti} = \phi^{\text{tot}} \int f(E)\sigma_i(E)dE. \tag{4.11}$$

The integral is the spectrum-averaged, or effective, cross section,

$$< \sigma_i >= \int f(E)\sigma_i(E)dE, \tag{4.12}$$

with the normalization

$$\int f(E)dE = 1. \tag{4.13}$$

This gives the compact expression

$$R_{Ti} = \phi^{\text{tot}} < \sigma_i > . \tag{4.14}$$

Similarly, the total reaction constant of the target, in [s^{-1}], is written as

$$R_T = \phi^{\text{tot}}(< \sigma_{non} > - < \sigma_{in} >), \tag{4.15}$$

where $< \sigma_{non} >$ is the spectrum-averaged non-elastic cross section and $< \sigma_{in} >$ is the spectrum-averaged inelastic cross section. The difference $< \sigma_{non} > - < \sigma_{in} >$ represents the probability to form a nuclide different from the original target nuclide in the nuclear reaction.

As discussed below, charged-particle irradiation requires a different expression, because the relevant spectrum is generated inside the target.

For typical irradiation sources, such as a research reactor or an accelerator, the total flux is of the order of 10^{12}–10^{14} particles [cm^{-2}s^{-1}]. Since σ has to be expressed in units of 10^{-24} cm^2 for one barn, absolute values of R are generally small. For cross sections in the range from 1 mb to 1 barn, one often obtains reaction constants of the order of $R \sim 10^{-10}$–10^{-13} s^{-1}.

The reaction constant depends on the intensity of the incident particle source, not on the irradiated target volume. It can therefore be compared consistently between different source types and with the decay constant: the higher the particle flux, the larger the production of the residual isotope relative to its decay. In the production and decay equations (4.2), the reaction constants compete with radioactive decay constants, with typical values of $\lambda \sim 10^{-2}$–10^{-4} s^{-1} for short-lived nuclides and $\lambda \sim 10^{-8}$–10^{-12} s^{-1} for long-lived nuclides.

4.2.1 Neutron irradiation

Medical isotope production by neutron reactions generally takes place in research reactors, including reactors dedicated to isotope production. Other promising neutron sources exist, for example sources with spectra peaked around 14 MeV, but these are used on a smaller scale or are still under development. In ISOTOPIA we provide access to the estimated neutron spectra that, to our knowledge, are available. Over the years, many neutron spectra have been compiled in the CONDERC database for research reactors, power reactors, fusion designs and related systems. Inserting such a normalized spectrum in Eq. (4.12) and providing the total flux in Eq. (4.14) leads directly to the production rate.

There are usually two refinements which reduce the calculated activity relative to Eq. (4.11):

- The flux may not be homogeneous, for instance because of neutron-absorbing materials in neighboring irradiation positions in the reactor core. In such cases, irradiation-specific estimates of the neutron spectrum have to be made, e.g. with a code such as MCNP.
- Self-shielding reduces the effective reaction rates in the production and depletion equations, because the flux inside the target is lower and spectrally different from the externally specified flux.

For the first issue, it is difficult to give a general prescription for a specific irradiation condition. We therefore assume here that the flux is the same throughout the relevant irradiation position.

Self shielding

A reasonable estimate can be given for self-shielding. Equation (4.11) is then replaced by

$$R_{Ti} = \phi^{\mathrm{tot}} \int G(E) f(E) \sigma_i(E) dE \tag{4.16}$$

where the self-shielding factor is

$$G(E) = \frac{1 - e^{-\Sigma(E).x}}{\Sigma(E).x}. \tag{4.17}$$

Here x is the material thickness and Σ is the macroscopic cross section,

$$\Sigma(E) = n_T \sigma_{att}, \tag{4.18}$$

for which we take the attenuation cross section σ_{att} to be the total cross section σ_{tot}. To assess the overall impact of self-shielding, it is useful to define the spectrum-averaged self-shielding factor

$$G_{av} = \int G(E) f(E) dE. \tag{4.19}$$

4.2.2 Photon irradiation

Photonuclear reactions are often induced by Bremsstrahlung spectra. Models for such spectra range from simple analytical expressions to spectra produced by full Monte Carlo simulations with codes such as GEANT4. In the present version we use a thick-target Kramers spectrum for $f(E)$ in Eq. (4.11).

4.2.3 Charged-particle irradiation

For charged-particle reactions, the expression for the reaction constant differs from Eq. (4.11), because the projectile spectrum is generated inside the target:

$$R_{Ti} = n_p \int_{E_{\text{back}}}^{E_{\text{beam}}} \frac{\sigma_i(E)}{S(E)} dE. \tag{4.20}$$

Here E_{beam} is the incident beam energy and E_{back} is the average particle energy leaving the back side of the target. As the projectiles traverse the target, their average energy decreases. The energy loss in the target is determined by the stopping power S in [MeV/cm], discussed below. The projectile rate density n_p, in [particles/(cm^{3}s)], is

$$n_p = N_{\text{proj}}/V_{\text{tar}}, \tag{4.21}$$

where V_{tar} is the active target volume in [cm^3], and N_{proj} is the number of projectiles impinging on the target per second,

$$N_{\text{proj}} = I_{\text{beam}}/(z_p q_e), \tag{4.22}$$

with I_{beam} the beam current in [A], z_p the projectile charge number and q_e the elementary charge. The effective path length in [cm] is

$$L_{\text{eff}} = \int_{E_{\text{back}}}^{E_{\text{beam}}} \frac{1}{S(E)} dE. \tag{4.23}$$

The integration limits E_{beam} and E_{back} determine the effective path length, and vice versa. The target volume is then given by the product of the beam surface S_{beam} in [cm^2] and the effective target length,

$$V_{tar} = S_{beam} L_{\text{eff}}. \tag{4.24}$$

Equation (4.20) can be rewritten in the form of the general expression for R, Eq. (4.11), by defining the spectrum function

$$g(E) = \frac{1/S(E)}{\int_{E_{\text{back}}}^{E_{\text{beam}}} \frac{1}{S(E)} dE}, \tag{4.25}$$

with

$$\int g(E) dE = 1. \tag{4.26}$$

One then obtains

$$R_{Ti} = \frac{N_{proj}}{V_{tar}} L_{\text{eff}} \int g(E)\sigma_i(E) dE \tag{4.27}$$

or

$$R_{Ti} = \frac{N_{proj}}{V_{tar}} L_{\text{eff}} < \sigma_i >_S, \tag{4.28}$$

where the subscript S indicates that the cross section is averaged over the spectrum associated with the stopping power.

Stopping power

The stopping power describes the average energy loss of projectiles in the target through atomic collisions as a function of projectile energy, in [MeV/cm]. We use the Bethe-Bloch formula [7]:

$$S(E) = \frac{dE}{dx} = 0.3070\,\rho\,\frac{Z}{A}\,\frac{z_p^2}{\beta^2}\,[ln\left(\frac{W_{max}^2}{I^2}\right) - 2\beta^2],\ \ \mathrm{MeV/cm} \tag{4.29}$$

with Z the target charge number and z_p the projectile charge number, while β represents a beam particle with relative velocity

$$\beta = \frac{v}{c} = \sqrt{\frac{E_{beam}(E_{beam} + 2m_0c^2)}{(E_{beam} + m_0c^2)^2}}, \tag{4.30}$$

where m_0 is the projectile rest mass. Furthermore, m_e is the electron mass and $\gamma = 1/\sqrt{1-\beta^2}$. The maximum energy transfer in a single collision, W_{max}, is

$$W_{max} = 2m_ec^2(\beta\gamma)^2. \tag{4.31}$$

For the mean excitation potential I, a semi-empirical expression is adopted:

$$\frac{I}{Z} = 9.76 + 58.8Z^{-1.19}\ \ \mathrm{eV}. \tag{4.32}$$

This expression is usually stated to be tested for $Z \geq 13$, but in the present implementation it is used for all values of Z.

Nuclear reaction and decay rates

With the definitions above, all ingredients are available to calculate the nuclide inventory $N_i(t)$ at any time during or after irradiation, except for the residual-production cross sections and radioactive decay data. These nuclear data are treated in a separate chapter.

When the target volume is included, one can also define reaction rates. The total production rate of nuclide j, in [atoms/s], is obtained by multiplying the reaction constant by the number of target atoms,

$$P_{Tj}(t) = R_{Tj}N_T(t). \tag{4.33}$$

The corresponding decay rate is the activity introduced below.

4.3 Activities

With $N_i(t)$ known, the activity A_i of the produced nuclide i as a function of irradiation time is

$$A_i(t) = \lambda_i\,N_i(t). \tag{4.34}$$

The activity is usually expressed in GBq or MBq. If the half-life of the produced isotope is long compared with the irradiation time, the expression for $N_i(t)$ simplifies to

$$N_i(t) = N_0\,R_{Ti}\,t, \tag{4.35}$$

and hence

$$A_i(t) = \lambda_i\,N_0\,R_{Ti}\,t. \tag{4.36}$$

In that case, the yield scales linearly with irradiation time t and reaction constant R_{Ti}. The production yield in, for example, [GBq/mAh] for accelerators, or the average production rate in [GBq/day] for reactors, is then a meaningful quantity.

4.4 Possible refinements

The formalism and ingredients specified above provide a reasonable-to-good estimate of isotope production, which is the first aim of ISOTOPIA. Several improvements are nevertheless possible and may lead to more accurate estimates. For all irradiation sources these are:

- Improved cross sections. Their uncertainties propagate directly to the estimated yields.
- More accurate decay data. In the present version, the JEFF radioactive decay data library is used. A combination of the best available decay data libraries may improve the estimated yields and impurity inventories.

For neutron sources in particular, possible refinements are

- a reactor-geometry dependent flux, in which the effect of more or less absorbing irradiation positions in the reactor is taken into account; this naturally leads to a user-specific reactor spectrum,
- the detailed geometry of the irradiated target and the associated consequences for self-shielding.

For photon sources, possible refinements are

- a better model for the Bremsstrahlung spectrum,
- a better estimate of the fraction of photons, after conversion from the electron beam, that are available for target irradiation,
- photon attenuation.

For charged-particle sources, an obvious refinement is

- a more detailed model for the stopping power.

5. Nuclear data libraries

In the previous chapter it was shown that the simulation of isotope production is determined by a small number of ingredients:

- the solution of the production and decay equations, Eq. (4.2),
- depending on the activation method:
  - for charged-particle induced activation, a model for the stopping power, in this work Eq. (4.29),
  - for photonuclear activation, a model for the Bremsstrahlung spectrum,
  - for neutron-induced activation, a model for the neutron spectrum,
- a radioactive decay data library,
- a cross section data library.

The first two ingredients define the formalism and the irradiation conditions. The last two ingredients determine the nuclear physics content of the calculation and therefore largely set the attainable predictive quality.

5.1 Radioactive decay data library

For radioactive decay data we use JEFF-3.3-RD. In the current version of ISOTOPIA only a limited subset of this information is required: the half-life and the basic decay mode along the decay chain, in particular whether electron or positron emission is involved. This is sufficient for the present activation formalism, but it also makes clear where future refinements of decay branching and isomer feeding could be introduced.

5.2 Cross section data library

The nuclear reaction data required by ISOTOPIA are, in essence, activation cross sections. TENDL is a complete nuclear data library in the sense that it contains cross sections for all relevant projectile, target, product and energy combinations within its range of applicability. For any projectile on any target, residual-production cross sections are available for all products that can be formed up to incident energies of 200 MeV.

For incident charged particles, TENDL has been produced automatically, with adjusted nuclear model parameters when experimental data are available. As a result, TENDL generally gives a reasonable-to-good representation of experimental excitation functions, and statistical analyses of its overall quality, also in comparison with other high-energy libraries, are available. At the same time, one should not pretend that a global library is always the best possible description of every individual reaction channel. For a limited but important set of medical isotope reactions, dedicated evaluations exist, and these have been incorporated in the medical isotope data library used here.

5.2.1 Neutrons

For neutron-induced reactions we use TENDL-2025 in pointwise format. This means that the neutron resonance parameters have been reconstructed into cross sections on a sufficiently fine energy grid.

5.2.2 Photonuclear reactions

For photonuclear reactions we use TENDL-2023. This choice was made because the basic settings for photonuclear reactions were found to be more suitable in the 2023 version than in the 2025 version.

5.2.3 Charged particles

The charged-particle cross section database is a merger of two sources:

- TENDL-2025 [1],
- the IAEA medical isotope database [3, 4, 5, 6]. This database has been produced through a series of IAEA collaborations and provides recommended cross sections for about 150 important medical isotope production reactions induced by charged particles. For these cases the experimental data were considered sufficiently abundant and/or of sufficiently high quality that empirical Padé fits could be used for the final evaluation, without relying on a nuclear model code. This gives excitation functions for restricted, but often most relevant, nuclide and energy ranges.

For charged particles, the medical isotope data library of ISOTOPIA is obtained by smoothly merging these two sources. TENDL is used for all projectiles, targets, products, reaction channels and energy ranges not covered by the IAEA database. For the approximately 150 IAEA reaction channels, the IAEA evaluation is extended outside its tabulated energy range by normalizing TENDL to the first and last energy points of the IAEA excitation function. In many cases, though not always, the IAEA energy interval contains the most relevant production range. For new or less conventional production routes only the TENDL cross sections are available.

Even for the well-known reaction channels for which a nuclear model estimate may not be needed for the primary product, the agreement of TALYS or TENDL with the measured excitation function remains important. The better the nuclear model reproduces the measured data, or the corresponding Padé fit, the more confidence one may have in the calculated production of other residual nuclides from the same irradiation, which may be the relevant impurities.

The combination of the IAEA database and TENDL currently provides what is likely the most complete medical isotope production library available for the purposes of ISOTOPIA. It can still be improved, in the usual way, by adjusting TALYS nuclear model parameters to reproduce experimental data more accurately and by performing further nuclide-by-nuclide evaluations.

5.3 Creation of medical isotope data library

The nuclear data library used by ISOTOPIA is called *isotopia.libs*. For future reference, we describe here how it is created.

- First, the raw data files from the IAEA medical isotope database are renamed to filenames consistent with the broader TALYS library system. For example, the ^{100}Mo(p,2n)^{99m}Tc cross section is stored as `p-Mo100-rp043099m.iaea.med`. This is a manual or semi-automated step.
- The file headers are then renamed to bring them in line with the rest of the *libraries/* database. For this we use the script `~/isotopia/misc/medrename.bash`.
- To further unify the database, every IAEA file is extended up to 200 MeV using a smooth transition to TENDL. This is done with `~/isotopia/misc/medmerge.bash`.
- There are several historical collections of IAEA files, for therapeutic, diagnostic and monitor reactions, among others. They all receive the same treatment, and older files are overwritten by newer files when applicable.
- In the directory *isotopia.libs* there is a script *create.isotopia.libs*. This script merges the latest TENDL version with the collection of specific IAEA medical isotope files. In practice, each reaction file from TENDL is copied, and the files that exist in the IAEA database overwrite the corresponding TENDL files. For completeness, all available EXFOR data are included as well.

This procedure produces a data library *isotopia.libs* for which the highest level consists of the incident particles,

```
p/
d/
t/
h/
a/
```

The next level contains the target nuclides,

```
....
p/Mo097/
p/Mo098/
p/Mo100/
etc.
```

The following level contains the evaluated nuclear data library and EXFOR data,

```
p/Mo100/exfor/
p/Mo100/iaea.2024/
```

All relevant EXFOR data available in computational XC4 format are stored in the `exfor/` directory. For medical isotope production only two subdirectories are needed:

```
p/Mo100/exfor/xs      : the channel cross sections, in particular the non-elastic
p/Mo100/exfor/residual: all residual production cross sections
```

The *iaea* directory contains the evaluated nuclear data tabulated per reaction channel in simple tables. For consistency with the *libraries/* database used in other projects, the same directory structure has been retained. For *isotopia.libs*, this gives for example

```
p/Mo100/iaea.2024/tables/xs      : the non-elastic cross section p-Mo100-MT005.iaea.2024
p/Mo100/iaea.2024/tables/residual: all residual prod. c.s. e.g. p-Mo100-rp043099m.iaea.2024
```

Finally, as an example of a file with residual production cross sections, *p-Mo100-rp043099m.iaea.2024* has the following structure:

```
# header:
#   title: Mo100(p,x)Tc99m cross section
#   source: IAEA medical isotope consortium
#   user: Arjan Koning
#   date: 2024-10-06
#   format: YANDF-0.2
# endf:
#   library: iaea.2024 + TENDL-2023
#   author: IAEA medical isotope consortium
#   year: 2024
# target:
#   Z: 42
#   A: 100
#   nuclide: Mo100
# reaction:
#   type: (p,x)
#   ENDF_MF: 6
#   ENDF_MT: 5
# residual:
#   Z: 43
#   A: 99
#   nuclide: Tc99m
#   level:
#     isomer: 1
# datablock:
#   quantity: cross section
#   columns: 3
#   entries:     457
##       E              xs             dxs
##     [MeV]           [mb]            [mb]
   6.000000E+00   1.100000E+00   1.100000E+00
   6.100000E+00   9.000000E-01   9.000000E-01
   6.200000E+00   8.000000E-01   8.000000E-01
   6.300000E+00   8.000000E-01   8.000000E-01
..................
```

This particular example contains data from the IAEA evaluation, extended with TENDL. By far the majority of files come directly from TENDL, as illustrated by *p-Mo100-rp043098.iaea.2024*:

```
# header:
#   title: Mo100(p,x)Tc98 cross section
#   source: ENDF
#   user: Arjan Koning
#   date: 2024-10-07
#   format: YANDF-0.2
# endf:
#   library: tendl.2023
#   author: A.J. Koning
#   year: 2023
# target:
#   Z: 42
#   A: 100
#   nuclide: Mo100
# reaction:
#   type: (p,x)
#   ENDF_MF: 6
#   ENDF_MT: 5
# residual:
#   Z: 43
#   A: 98
#   nuclide: Tc98
# datablock:
#   quantity: cross section
#   columns: 2
#   entries: 31
##       E             xs
##     [MeV]          [mb]
   1.685404E+01   0.000000E+00
   1.700000E+01   6.829083E-02
   1.800000E+01   3.388542E+01
   1.900000E+01   1.525024E+02
..................
```

In summary, TENDL is used for the energy ranges outside the IAEA evaluations and for all isotopes and reaction products not covered by those evaluations.

6. Reference Guide

In this part, all ISOTOPIA keywords will be described, one per page. The description of each keyword is as follows:

- Name of the keyword
- Explanation
- Examples
- Range of allowed values
- Default value
- Comments (optional), when we feel that some extra warnings or explanation for proper use is appropriate.

In principle, all keywords may be referred to in the other parts of this tutorial.

6.1 ISOTOPIA keywords

As explained above, the minimum input file has only a few keywords, and leaves all choices for the other parameters to the author. In general, you probably want to be more specific. Below, we will explain all the possible keywords. For each keyword, we give an explanation, a few examples, the default value, and the theoretically allowed numerical range. Remember that you can always find all adopted default values for all parameters at the top of the standard *output* file of ISOTOPIA.

abundance

File with tabulated abundances. The **abundance** keyword is only active for the case of a natural target, i.e. if **mass 0**. By default, the isotopic abundances are read from the *isotopia/files/abundance/* database. It can however be imagined that one wants to analyze production from targets of a certain isotopic enrichment. On the input line, we read **abundance** and the filename. From each line of the file, TALYS reads *Z*, *A* and the isotopic abundance with the format (2i4,f11.6). An example of an abundance file, e.g. *abnew*, different from that of the database, is

```
82 206  24.100000
82 207  22.100000
82 208  52.400000
```

where we have left out the "unimportant" ^{204}Pb (1.4%). ISOTOPIA automatically normalizes the abundances to a sum of 1, leading in the above case to 24.44 % of ^{206}Pb, 22.41 % of ^{207}Pb and 53.14 % of ^{208}Pb in the actual calculation.

Examples

abundance abnew

Range

abundance can be equal to any filename, provided it is present in the working directory.

Default

If **abundance** is not given in the input file, abundances are taken from *files/abundance* and calculations for all isotopes are performed.

Adepth

The depth, relative to the target mass number, to which isotopes are considered for cross sections that contribute to the production path. For example if **Adepth 5** for the p + ^{100}Mo reaction, only yields for elements defined by **Zdepth** and isotopes with mass numbers between 95 and 101 will be calculated.

Examples

Adepth 2
Adepth 10

Range

0 $\leq$ **Adepth** $\leq$ **Atarget - 1** where Atarget is the mass number of the target.

Default

Adepth 20.

Area

Area of the target in cm^2.

Examples

Area 10.

Range

0 $<$ **Area** $<=$ **10000**.

Default

Area 1. cm^2.

crosspath

Path for the cross section database. You should have this path hardwired in subroutine *machine.f*, but it may be helpful to easily change between different versions of the cross section database.

Examples

crosspath /home/koning/libraries
crosspath /home/qaim/tendl.2019

Range

crosspath should exist.

Default

Default: **crosspath ∼/isotopia.libs/**

currentunit

Unit for accelerator current.

Examples

currentunit mua: micro-Ampere
currentunit ma: milli-Ampere
currentunit A: Ampere

Range

currentunit should be equal to one of the units above.

Default

currentunit ma.

decay

Flag to include decay from parent nuclide. Basically used for code diagnostics.

Examples

decay y
decay n

Range

y or **n**

Default

decay y

Eback

For charged particle irradiation only: The lower end of the energy range at the back end of the target in MeV. This energy degradation is directly related to the effective thickness of the target.

Examples

Eback 130.
Eback 12.

Range

0. $\leq$ **Eback** $<$ **Ebeam** and 0. $\leq$ **Eback** $<$ 250. MeV

Default

Eback = Ebeam - 5 MeV.

Ebeam

For charged particle irradiation only: The incident energy of the particle beam in MeV.

Examples

Ebeam 140.
Ebeam 16.

Range

$0 <$ **Ebeam** $<= 250.$ MeV.

Default

None.

element

Either the nuclear symbol or the charge number Z of the target nucleus can be given. Possible values for **element** range from Li (3) to C4 (124).

Examples

element pu
element 41
element V
element B9

Range

1 $\leq$ **element** $\leq$ **124** or **H** $\leq$ **element** $\leq$ **C4**.

Default

None.

Comments

- To accommodate target nuclides with $Z > 110$ the element names are defined as follows: Rg(111), Cn(112), Nh(113), Fl(114), Mc(115), Lv(116), Ts(117), Og(118), B9(119), C0-4(120-124), which takes as a basis for $Z = 100$ the symbol $A0$, etc. unless an official name has been assigned to it, which is currently the case for $Z \leq 118$. Clearly, if we ever need or wish to go beyond $Z = 124$, there are enough symbols left. Obviously the symbols for Z above 118 will be changed as soon as official names are assigned to them.

fluxtotal

For neutrons: Total particle flux in [cm^{-}2s^{-}1].

Examples

fluxtotal 2.e14
fluxtotal 5.e13

Range

0 $<$ **fluxtotal** $<=$ **1e38**.

Default

fluxtotal 1.e14.

fgamma

Electron-to-photon conversion efficiency, in photons/electron, for Bremsstrahlung.

Examples

fgamma 0.2

Range

$\mathbf{0} < \textbf{fgamma} <= \mathbf{1}$.

Default

fgamma 0.3.

format

The format of the metadata in the output files. It may be helpful to specify the version. You may edit the 'format' variable variable in input.f90 and put your own version there. However, you can also use the input keyword.

Examples

format YANDF-0.5beta2

Range

Any string

Default

YANDF-0.4

Ibeam

For charged particles and photons: Particle beam current in the unit of **currentunit** (which has default mA).

Examples

Ibeam 0.1
Ibeam 5.

Range

0 < **Ibeam** <= **10000**.

Default

Ibeam 1.

mass

The target mass number A. The case of a natural element can be specified by **mass 0**. Then, an ISOTOPIA calculation for each naturally occurring isotope will be performed (see also the **abundance** keyword, p. 32), after which the results will be properly weighted and summed.

Examples

mass 112
mass 0

Range

mass 0 or $\mathbf{5} <$ **mass** $\leq \mathbf{339}$. The extra condition is that the target nucleus, i.e. the combination of the target **mass** and **element** must be present in the cross section database, which corresponds to all nuclei with a half life longer than 1 day.

Default

None.

massunit

Unit for isotope yield (to give relative production rate) or mass unit, in weight.

Examples

massunit mug: microgram
massunit mg: milligram
massunit g: gram
massunit kg: kilogram

Range

massunit should be equal to one of the units above.

Default

massunit g.

outcross

Flag to output cross sections to separate files.

Examples

outcross y
outcross n

Range

y or **n**

Default

outcross n

projectile

Seven different symbols can be given as **projectile**, namely **n, g, p, d, t, h, a** representing neutron, gamma (photon), proton, deuteron, triton, ^{3}He, and alpha, respectively.

Examples

projectile n
projectile g
projectile p
projectile d

Range

projectile must be equal to **n, g, p, d, t, h, a**.

Default

None.

radiounit

Unit for radioactivity, to be used in the output files for isotope production.

Examples

radiounit Bq: Becquerel
radiounit kBq: kiloBecquerel
radiounit MBq: MegaBecquerel
radiounit GBq: GigaBecquerel
radiounit Ci: Curie
radiounit mCi: milliCurie
radiounit kCi: kiloCurie

Range

radiounit should be equal to one of the units above.

Default

radiounit Bq.

rho

Material density of the target in g/cm^3.

Examples

rho 10.

Range

0 $<$ **rho** $<=$ **100**.

Default

rho is read from a hardwired material density table.

selfshield

For neutron sources: flag to take into account self shielding.

Examples

selfshield y
selfshield n

Range

y or **n**

Default

selfshield y

source

The source of the calculations. obviously this is ISOTOPIA, but it may be helpful to specify the version such that it appears in the metadata of the many output files. You may edit the 'source' variable variable in input.f90 and put your own version there. However, you can also use the input keyword.

Examples

source ISOTOPIAbeta3v20260223

Range

Any string

Default

ISOTOPIA

targetmass

For neutrons or photons: Mass of the target in the **massunit** unit (default: g).

Examples

targetmass 0.1
targetmass 5.

Range

0 $<$ **targetmass** $<=$ **1e9**.

Default

targetmass is not used but calculated from **rho** and volume of target.

thickness

For neutrons or photons: thickness of the target in cm. For protons this is calculated from the stopping power.

Examples

thickness 0.1
thickness 5.

Range

0 $<$ **thickness** $<=$ **10000** cm.

Default

thickness 0.05 cm.

timeunit

Unit for time for irradiation, cooling or relative production rate.

Examples

timeunit s: second
timeunit h: hour
timeunit d: day

Range

timeunit should be equal to one of the units above.

Default

timeunit h.

Tirrad

Irradiation time. A general input for the irradiation time has been enabled. On the input line we read integer values and time units, which can be y (years), d (days), h (hours), m (minutes) or s (seconds). These all need to be separated by blanks.

Examples

Tirrad	**2 d 5 h**
Tirrad	**32 h 30 m**
Tirrad	**1 d 6 h 24 m 12 s**

Range

0 $<=$ **Tirrad** $<=$ **1e6** for every time unit.

Default

Tirrad 1 d.

Tcool

Target cooling time. A general input for the cooling time has been enabled. On the input line we read integer values and time units, which can be y (years), d (days), h (hours), m (minutes) or s (seconds). These all need to be separated by blanks.

Examples

Tcool	**2 d 5 h**
Tcool	**32 h 30 m**
Tcool	**1 d 6 m 24 m 12 s**

Range

0 <= **Tcool** <= **1e6** for every time unit.

Default

Tcool 1 d.

user

The current user who produces the result. You may edit the 'user' variable in input.f90 and put your own name there. It will appear in the metadata of the various output files. However, you can also use the input keyword.

Examples

user Vlad Avrigeanu

Range

Any string

Default

Arjan Koning

xsfile

Local file with cross sections. With this keyword, residual production cross sections from TENDL can be overruled with values given in your working directory. On the input line, we read **xsfile**, Z and A of the residual product, the filename, and optionally the number of the produced isomer. This file should consist of Energy [MeV] - cross section [mb] values in x-y form, with a similar structure as those of the cross section database.

Examples

xsfile 25 56 Mn56.exp
xsfile 42 100 Mo100.loc 1

Range

xsfile should exist.

Default

None.

yieldunit

Unit for isotope yield (to give relative production rate) or mass unit, in weight.

Examples

yieldunit mug: microgram
yieldunit mg: milligram
yieldunit g: gram
yieldunit kg: kilogram

Range

yieldunit should be equal to one of the units above.

Default

yieldunit g.

ZAoutput

Flag to use numerical Z and A values in the filenames instead of nuclide names.

Examples

ZAoutput y
ZAoutput n

Range

y or **n**

Default

ZAoutput n

Zdepth

The depth, relative to the target charge number, to which isotopes are considered for cross sections that contribute to the production path. For example if **Zdepth 2** for the p + Mo reaction, only yields for Tc, Mo, Nb and Zr isotopes will be calculated.

Examples

Zdepth 2
Zdepth 10

Range

0 $\leq$ **Zdepth** $\leq$ **Ztarget - 1** where Ztarget is the charge number of the target.

Default

Zdepth 10.

Table 6.1: The keywords of ISOTOPIA.

Keyword	Range	Default	Page
abundance	filename	no default	32
Adepth	0-Atarget	20	33
Area	0. - 10000.	1.	34
crosspath	directory name	~/isotopia.libs/	35
currentunit	muA, mA, or A	mA	36
decay	y,n	y	37
Eback	1.e-11 - 250.	Ebeam - 5	38
Ebeam	1.e-11 - 250.	None	39
element	3 - 124 or Li - C4	None	40
fgamma	0. - 1.	0.3	42
fluxtotal	0. - 1.e38	1.e14	41
format	format name	YANDF-0.4	43
Ibeam	0. - 10000.	1.	44
mass	0, 5 - 339	None	45
massunit	mug, mg, g or kg	g	46
outcross	y,n	n	47
projectile	n,g,p,d,t,h,a,	None	48
radiounit	(k,M,G)Bq, (m,k)Ci	Bq	49
rho	0. - 100.	None	50
selfshield	y,n	y	51
source	name of source	ISOTOPIA	52
targetmass	0. - 1.e9	None	53
thickness	0. - 10000.	0.05	54
timeunit	s, h or d	h	55
Tirrad	0 - 1.e6	1 d	56
Tcool	0 - 1.e6	1 d	57
user	your name	Arjan Koning	58
xsfile	file name	none	59
yieldunit	mug, mg, g or kg	g	60
ZAoutput	y,n	n	61
Zdepth	0-Ztarget	10	62

7. Sample cases

The purpose of ISOTOPIA is that every imaginable production route for a medical isotope can be simulated. To strengthen this statement, we will discuss a few sample cases. In each case, the ISOTOPIA input file, the relevant output and if appropriate, a figure will be presented. The description of the first sample case is the longest, since the output of ISOTOPIA will be discussed in complete detail. Obviously, that output description is also applicable to the other sample cases. The entire collection of sample cases serves as (a) verification of ISOTOPIA: the sample output files should coincide, apart from numerical differences of insignificant order, with the output files obtained by you, and (b) validation of ISOTOPIA: the results should be comparable to those obtained in other publications. The computation time of ISOTOPIA is very small. In general, we will explain the keywords again as they appear in the input files below. If not, consult Table 6.1, which will tell you where to find the explanation of a keyword.

7.1 Sample p-Mo100-Tc099m: Accelerator production of ^{99m}Tc

This sample case considers one of the most discussed alternatives for reactor-produced ^{99}Mo: the direct production of ^{99m}Tc from ^{100}Mo with a proton accelerator via the (p,2n) reaction. The input file for this case is as follows,

```
projectile p
element Mo
mass 100
Ebeam 16.
Eback 12.
Ibeam 0.15
```

Running this input file with

isotopia < isotopia.inp > isotopia.out

gives an output file with the general characteristics of the reaction. The output file starts with a display of the version of ISOTOPIA you are using, the name of the author, the Copyright statement and then an exact copy of the input file as given by the user is returned.

```
    ISOTOPIA-2.1    (Version: December 29, 2024)

           Prediction of medical isotope production with accelerators

 Copyright (C) 2024  A.J. Koning

 User: Arjan Koning
 Date: 2024-12-22

 ########## USER INPUT ##########

 USER INPUT FILE

 projectile p
 element mo
 mass 100
 ebeam 16.
 eback 12.
 ibeam 0.15

 USER INPUT FILE + DEFAULTS

 Keyword           Value   Variable     Explanation

 user      Arjan Koning    user         user for this calculation
 source    ISOTOPIA        source       source for this calculation
 format    YANDF-0.2       oformat      format for output
 #
 # Nuclear reaction
 #
 projectile          p     ptype0       type of incident particle
 element            Mo     Starget      symbol of target nucleus
 mass              100     Atarget      mass number of target nucleus
 #
 # Accelerator
 #
 Ebeam              16.000 Ebeam        incident energy in MeV
 Eback              12.000 Eback        lower end of energy range in MeV
 Ibeam               0.150 Ibeam        beam current in mA
 Tirrad              1     Tirrad       d of irradiation time
 radiounit             gbq radiounit    unit for radioactivity
 yieldunit             num yieldunit    unit for isotope yield
 #
 # Target
 #
```

```
 Area                1.000 Area         target area in cm^2
 Tcool               1     Tcool        d of cooling time
 rho                -1.000 rhotarget    target density [g/cm^3]
 #
 # Cross section and decay data
 #
 Zdepth             10     Zdepth       depth to which Z numbers are scanned for cross sections
 Adepth             20     Adepth       depth to which A numbers are scanned for cross sections
```

Not only the keywords that you have specified in the input file are listed, but also all the defaults that are set automatically. The corresponding Fortran variables are also printed, together with a short explanation of their meaning. This table can be helpful as a guide to change further input parameters for a next run. You may also copy and paste the block directly into your next input file.

In the next output block we print the main parameters that characterize the production process. Note that some of these have been given in the input, while others are automatically calculated.

```
 Summary of isotope production for p + Mo100

 Maximal irradiation time:       0 years   1 days  0 hours  0 minutes  0 seconds
 Cooling time:       0 years   1 days  0 hours  0 minutes  0 seconds
 E-beam [MeV]:   1.600000E+01
 E-back [MeV]:   1.200000E+01
 Beam current [mA]:        0.150 mA
 Target material density [g/cm^3]:   1.022000E+01
 Target area [cm^2]:   1.000000E+00
 Effective target thickness [cm]:   2.097696E-02
 Effective target volume [cm^3]:   2.097696E-02
 Effective target mass [g]:   2.143845E-01
 Number of target atoms:    1.291053E+21
 Number of incident particles [s^-1]:   9.362260E+14
 Produced heat in target [kW]:   6.000000E-01
```

The block with final results is

```
 (Maximum) production and decay rates per isotope

 Total production rate [s^-1]:   9.624934E-10

#  Nuc     Production rate Decay rate     Activity       #isotopes      Yield          Isotopic fr
#             [s^-1]          [s^-1]         [GBq]          [   ]         [GBq/mAh]
 Tc 101     9.115965E-13   8.135531E-04   1.176901E+00   1.446618E+12   3.427384E-03   0.00004
 Tc 100     6.088757E-11   4.387007E-02   7.860895E+01   1.791858E+12   3.032753E-01   0.00051
 Tc  99     7.742050E-10   1.026401E-13   2.616365E+02   7.370476E+16   6.840031E-10   0.54037
 Tc  99 g   5.875616E-10   1.026401E-13   6.726828E-06   6.553800E+16   5.190665E-10   0.40869
 Tc  99 m   2.162431E-10   3.203675E-05   2.616365E+02   8.166760E+15   5.800662E-02   0.05093
 Mo 100     1.174375E-10   0.000000E+00   0.000000E+00   1.290946E+21   0.000000E+00   1.00000
..................
  End of ISOTOPIA calculation for p + Mo100
```

The time-dependent production of each isotope is found the files *ZZAAA.act* or *ZZAAAm.act* if is concerns isomeric production, with ZZ the charge number in (i2.2) format and AA the mass number in (i3.3) format.

For this case, obviously most important is the production of ^{99m}Tc, given in file *Tc099m.act*, which looks as follows,

```
# header:
#   title: Mo100(p,x)Tc99m Isotope production
#   source: ISOTOPIA
#   user: Arjan Koning
#   date: 2024-12-22
#   format: YANDF-0.2
# target:
#   Z: 42
#   A: 100
#   nuclide: Mo100
# reaction:
#   type: (p,x)
#   ENDF_MF: 6
#   ENDF_MT: 5
# residual:
#   Z: 43
#   A: 99
#   nuclide: Tc99m
# parameters:
#   Beam current [mA]:  1.500000E-01
#   E-Beam [MeV]:  1.600000E+01
#   E-Back [MeV]:  1.200000E+01
#   Initial production rate [s^-1]:  2.162431E-10
#   Decay rate [s^-1]:  3.203675E-05
#   Initial production yield [GBq/mAh]:  5.800662E-02
#   Total activity at EOI [GBq]:  2.616365E+02
#   Irradiation time:       0 years   1 days  0 hours  0 minutes  0 seconds
#   Cooling time:       0 years   1 days  0 hours  0 minutes  0 seconds
#   Half-life:       0 years   0 days  6 hours  0 minutes 36 seconds
#   Maximum production:       0 years   3 days 18 hours 17 minutes 18 seconds
# datablock:
#   quantity: Isotope production
#   columns: 6
#   entries: 100
##     Time         Activity       Isotopes         Yield     Isotopic_frac.      Time
##     [sec]          [GBq]           []          [GBq/mAh]         []              [h]
   1.728000E+03   1.503532E+01   4.693146E+14   5.800662E-02   1.336947E-01   4.800000E-01
   3.456000E+03   2.926088E+01   9.133534E+14   5.488257E-02   1.306088E-01   9.600000E-01
   5.184000E+03   4.272030E+01   1.333478E+15   5.192677E-02   1.275825E-01   1.440000E+00
   6.912000E+03   5.545484E+01   1.730975E+15   4.913017E-02   1.246355E-01   1.920000E+00
   8.640000E+03   6.750352E+01   2.107065E+15   4.648413E-02   1.217707E-01   2.400000E+00
.....................
```

In the file above, the general irradiation characteristics are repeated, as well as the total production rates. The activity as a function of time is presented in Fig. 7.1. Note that the production yield decreases as a function of time since ^{99m}Tc has a half life of 6 hours. Also the last column in this file can be related to the purity or specific activity of ^{99m}Tc relative to the other Tc isotopes and ^{99c}Tc.

Note that the directory also lists the following files

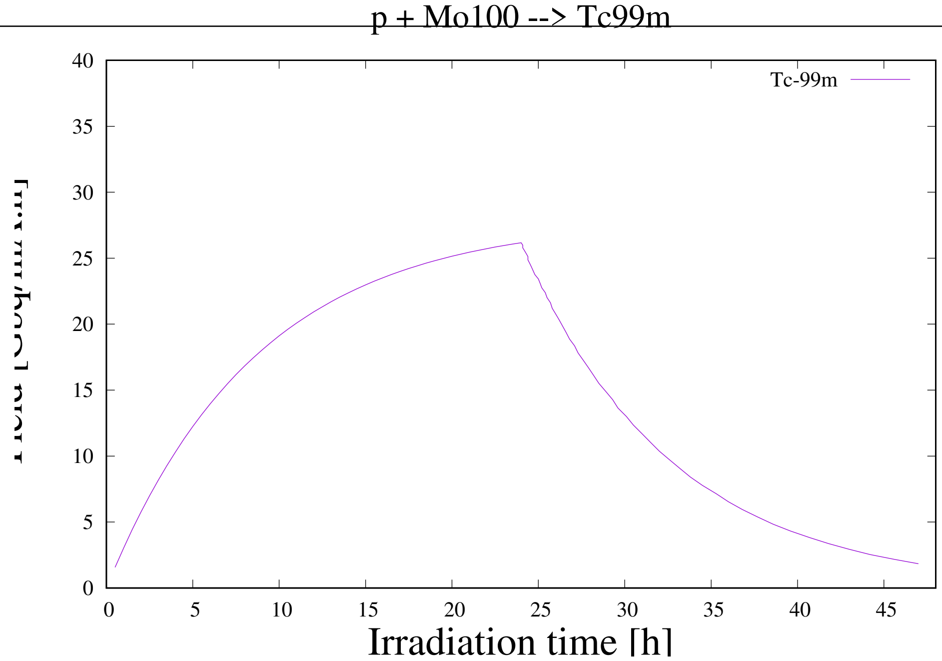

Figure 7.1: Production of Tc-99m

```
Mo099.act
Mo100.act
Tc099.act
Tc099g.act
Tc099m.act
Tc100.act
Tc101.act
```

which represent the isotopes, of which some can be regarded as 'impurities', which are produced simultaneously.

7.2 Sample a-Bi209-At211: alpha-induced production of therapeutic nuclide ^{211}At

This production route is listed in Table 1 of Ref. [8]. The corresponding input file for this case is

```
projectile a
element Bi
mass 209
Ebeam 28.
Eback 10.
Ibeam 0.100
```

Ref. [8] lists an initial production yield of 17.5 Mbq/(μA.h), while ISOTOPIA produces a value of 22.5, see the output file. Note that there is still some uncertainty with regards to the production cross section.

7.3 Sample p-Ga069-Ge068: Production of the ^{68}Ge generator for ^{68}Ga

There are various production routes for this isotope. The input file for this case is

```
projectile p
element Ga
mass 69
Ebeam 25.
Eback 15.
Tirrad 7 d
radiounit Ci
outcross y
```

Note that in this input file we have include some other features of ISOTOPIA, which of course can also be used in all other input files. The irradiation time is 7 days, while we request the output to be in Ci instead of GBq. In this case, the output file of most interest is probably *Ge068.act*. Also, all residual production cross sections for this reaction are printed in separate files.

7.4 Sample p-Th232-Ac225: Production of ^{225}Ac from ^{232}Th

As an example of the high-energy end of the domain of ISOTOPIA, we include the following sample case,

```
projectile p
element Th
mass 232
Ebeam 192
Eback 170.
Ibeam 0.10
```

Note that activation files for many isotopes are produced. The higher the incident energy, the more side-products are made, and many of them could be unwanted.

8. Outlook and conclusions

This tutorial describes ISOTOPIA, a software package for the prediction of medical isotope production with accelerators, photons and neutron sources. Its purpose is to provide a transparent calculational path from an irradiation scenario, through nuclear reaction and decay data, to the activities of the desired products and the associated impurities.

The formalism used to calculate isotope activities is based on direct production and radioactive decay of the produced nuclides. Feeding through β decay of other nuclei formed in the nuclear reaction is included for one parent nuclide, which is generally sufficient for activation calculations. In a fully general treatment one would also include all β-decay feeding paths, as well as nuclear reactions on already produced nuclei, which may become relevant at high target burn-up. For most cases of present interest this extra complexity is not necessary. Nevertheless, a future version of ISOTOPIA may be extended with a general differential-equation solver.

ISOTOPIA is complete in the sense that it produces results for all possible isotope production paths covered by its nuclear data library. The quality of the prediction is therefore directly related to the quality of the underlying cross sections. The backbone of the cross section database is TENDL-2025, with TENDL-2023 used for photons. TENDL has gradually evolved from a calculated data library into an evaluated data library. This means that, in addition to global theoretical improvements in the TALYS code, an increasing number of experimental cross section data sets are incorporated in the evaluation process. Until this is complete, and probably also afterwards, the IAEA medical isotope database remains indispensable for the approximately 150 important reaction channels for which it provides dedicated evaluations.

In terms of functionality, one obvious extension is the possibility to answer the “inverse question”: given a desired isotope, how do the possible production routes compare in terms of projectile type, target choice, energy range, yield, impurities and cooling behavior, and what is the optimal production route? The current version of ISOTOPIA treats the corresponding forward problem. Starting from an intuitively promising production route, it calculates the relevant quantities. To answer the inverse question, a loop over the full ISOTOPIA system has to be constructed, followed by an efficient search or optimization procedure. Given the short calculation time of a typical ISOTOPIA run, this appears feasible. The most robust approach may be to generate a complete database by running ISOTOPIA over all nuclides and projectiles, and then analyze that database for optimal production scenarios.

Bibliography

[1] A.J. Koning et al. "TENDL: Complete Nuclear Data Library for innovative Nuclear Science and Technology". In: *Nuclear Data Sheets* 155 (2019), p. 1.

[2] A.J. Koning, S. Hilaire, and S. Goriely. "TALYS: modeling of nuclear reactions". In: *European Physical Journal A* 59 (2023), p. 131.

[3] E. Betak et al. "Nuclear data for the production of therapeutic radionuclides". In: *IAEA Technical Reports Series, IAEA, Vienna, Austria (2011) ISBN 978-92-0-115010-3. editors S.M. Qaim, F. Tarkanyi, R. Capote, technical report STI/DOC/010/473* (2011).

[4] A. Hermanne et al. "Reference cross sections for charged-particle monitor reactions". In: *Nucl. Data Sheets* 148 (2018), pp. 338–382.

[5] F. T. Tarkanyi et al. "Recommended nuclear data for medical radioisotope production: diagnostic positron emitters". In: *J. Rad.Nucl.Chem.* 319 (2019), pp. 487–531.

[6] J. W. Engle et al. "Recommended Nuclear Data for the Production of Selected Therapeutic Radionuclides". In: *Nucl. Data Sheets* 155 (2019), pp. 56–74.

[7] W. R. Leo. "Techniques for Nuclear and Particle Physics Experiments". In: *Springer Verlag, Berlin* (1994).

[8] B. Scholten S.M. Qaim I. Spahn and B. Neumaier. "Uses of alpha particles, especially in nuclear reaction studies and medical radionuclide production". In: *Radiochimica Acta* 104 (2016), p. 601.

[9] H. Bateman. In: *Proc. Cambridge Phil. Soc.* 15 (1910), p. 423.

[10] W. Rubinson. In: *J. Chem. Phys.* 17 (1949), p. 542.

A. Derivation of the nuclide production equations

A.1 Differential equations for production and depletion of nuclides

The most general situation is that of the irradiation of a piece of material, consisting of many different nuclides, which were either present at the start of the irradiation, have been formed, or will be formed during the irradiation, by either the primary flux of particles, by secondary particles, or by radioactive decay. If we have a total of K different nuclides and the number of each nuclide k is N_k, then the temporal development of such a system, during irradiation and decay, is described by K differential equations:

$$\begin{aligned}
\frac{dN_1(t)}{dt} &= \sum_{k=1,k\neq 1}^{K} \Lambda_{k\rightarrow 1} N_k(t) - \Lambda_1 N_1(t) \\
&\dots \\
\frac{dN_i(t)}{dt} &= \sum_{k=1,k\neq i}^{K} \Lambda_{k\rightarrow i} N_k(t) - \Lambda_i N_i(t) \\
&\dots \\
\frac{dN_K(t)}{dt} &= \sum_{k=1,k\neq K}^{K} \Lambda_{k\rightarrow K} N_k(t) - \Lambda_K N_K(t)
\end{aligned} \tag{A.1}$$

In each equation, the first term is a feeding term. In the most general case, various parent nuclides k may contribute to the formation of nuclide i, hence the summation over k, with $\Lambda_{k\rightarrow i}$ the partial formation rate for any possible parent nuclide k to nuclide i. Each partial formation rate can be expressed as

$$\Lambda_{k\rightarrow i} = \lambda_{k\rightarrow i} + R_{k\rightarrow i} \tag{A.2}$$

with $\lambda_{k\rightarrow i}$ the (partial) radioactive decay rate and $R_{k\rightarrow i}$ the (partial) nuclear reaction rate for any possible parent nuclide k to nuclide i. The second terms of Eq. (A.1) are loss terms due to radioactive decay and nuclear reactions from nuclide i to any other nuclide. Here the total depletion rate (we interchange "depletion" with "formation" whenever appropriate) for nuclide i is

$$\Lambda_i = \lambda_i + R_i \tag{A.3}$$

where the total decay rate for nuclide i is

$$\lambda_i = \sum_{k=1,k\neq i}^{K} \lambda_{i\to k}$$

and the total nuclear reaction rate for nuclide i is

$$R_i = \sum_{k=1,k\neq i}^{K} R_{i\to k} \tag{A.4}$$

The entire loop over k may run over nuclides in their ground or isomeric states. Theoretically, the sum over reaction rates could include secondary particles (neutrons, photons, alpha particles etc.) formed after the first interaction of the incident beam with the material, over the entire outgoing energy spectrum. Since the number of nuclides i, $N_i(t)$ may appear simultaneously in many equations, due to its possible formation, or depletion, by many different nuclear reactions, it is clear that such a coupled system can only be solved by complicated mathematical and computational techniques. In fact, the most exact simulation would involve a Monte Carlo 3D transport calculation in which all primary and secondary particles are taken into account, including complete cross section libraries for all possible particles, coupled with an activation code that keeps track of the nuclide inventory. If we neglect such thick target transport issues, a system of equations like (A.1) is often solved by methods developed by Bateman [9] for radioactive decay and later generalized for source terms by Rubinson [10].

Fortunately, the situation is often not as complex as sketched above, since very reasonable approximations can be introduced into Eq. (A.1), which represent cases which are used in practice. First, let us start with the common case of the irradiation of a target which contains only one natural element at the start of the irradiation. This set of equations can be then be separated into a linear combination of contributions by each target *isotope*. Hence, we solve Eq. (A.1) one by one for each target isotope and add these contributions at the end to get the activation for the target material. Then, for such a mono-isotopic target T we have at the start of the irradiation

$$\begin{aligned} t=0: N_T &= N_T(0) = N_0 \\ N_i &= 0 \\ &\dots \\ N_K &= 0 \end{aligned} \tag{A.5}$$

Since in practice our target nuclide is not radioactive, the loss terms reduce to $\Lambda_T = R_T$ in Eq. (A.1) for N_T. Next, if a substantial part of the target is converted into other nuclides, beam particles may interact with atoms other than the original target atoms. However, this is only a concern for very long irradiation times. For most practical applications, like medical isotope production, we can often assume that the burn-up of the target is small and that the target composition does not change much during the irradiation (this will be confirmed by some of our sample cases). Hence, in addition we will assume there are no nuclear reaction or radioactive decay feeding terms for the *target* nuclide T.

This means that the temporal evolution of the target nuclide is described by the following simple equation

$$\frac{dN_T(t)}{dt} = -R_T N_T(t) \tag{A.6}$$

which basically states that the only thing that happens to the target material is burn-up through nuclear reactions.

For the nuclides that are produced during the irradiation, which are of course more interesting, we can make similar reasonable assumptions. We assume there is no loss of *produced* nuclides of interest through nuclear reactions with beam particles, i.e. we assume nuclides which are produced are not hit twice (again, this assumption becomes less accurate at very long irradiation times). This is equivalent to stating that the nuclides of interest are only produced from nuclear reactions on the target nuclides or from decay of other products formed during the irradiation. Also, consistent with the assumption for the target nuclide that nuclides are not hit twice, we have $\Lambda_i = \lambda_i$ for the depletion term, and assume that no other nuclear reactions lead to the nuclide of interest. Thus for the produced nuclides i we obtain

$$\frac{dN_i(t)}{dt} = \sum_{k=1,k\neq i}^{K} \lambda_{k\to i} N_k(t) + R_{T\to i} N_T(t) - \lambda_i N_i(t) \tag{A.7}$$

Nuclides often have only one or a few radioactive decay modes. Usually, only beta decay to the ground state or an isomer needs to be considered, although in some cases alpha decay may occur as well. If we neglect alpha decay, the summation in the first term above reduces to only one term. Thus, Eq. (A.7) reduces to

$$\frac{dN_i(t)}{dt} = \lambda_{p\to i} N_p(t) + R_{T\to i} N_T(t) - \lambda_i N_i(t) \tag{A.8}$$

where we now use the subscript p for the parent nuclide which decays to nuclide i. The parent nuclide itself is produced from nuclear reactions on the target and is described by the equation,

$$\frac{dN_p(t)}{dt} = R_{T\to p} N_T(t) - \lambda_p N_p(t) \tag{A.9}$$

where for simplicity we have left out a possible feeding term $\lambda_{g\to p} N_g(t)$ from its own parent (i.e. the grandparent g of nuclide i). It neglects radioactive decay to channels produced by multiple proton emission, e.g. ^{120}Te(p,2p)^{119}Sb and the contribution from its feeding channel ^{120}Te(p,2n)^{119}I + $2\beta^+$. Although the (p,2p) channel is generally smaller and therefore often not relevant for most practical medical isotope production routes, it would be safer that at higher energies feeding from radioactive decay is taken into account. For that a more general solution of Eq. (A.1) would have to be implemented, similar to solutions for neutron-induced problems where processes like e.g. multiple decay of fission fragments needs to be taken into account. For most charged-particle induced reactions of interest, the current approximation is justified.

A.2 Solution of the differential equations for production and depletion

Here we derive the solutions (4.2) of the differential equations (A.6)-(A.9). Eq. (4.2) has been implemented in ISOTOPIA.

For the target nuclide we have

$$\frac{dN_T(t)}{dt} = -R_T N_T(t). \tag{A.10}$$

This is easily integrated to give

$$N_T(t) = N_0\, e^{-R_T t}. \tag{A.11}$$

For every parent nuclide we have

$$\frac{dN_p(t)}{dt} = R_{T\to p} N_T(t) - \lambda_p N_p(t). \tag{A.12}$$

Multiplying both sides with $e^{\lambda_p t}$ gives

$$e^{\lambda_p t}\frac{dN_p(t)}{dt} = e^{\lambda_p t}\left[R_{T\to p} N_T(t) - \lambda_p N_p(t)\right], \tag{A.13}$$

or,

$$e^{\lambda_p t}\frac{dN_p(t)}{dt} + e^{\lambda_p t}\lambda_p N_p(t) = e^{\lambda_p t} R_{T\to p} N_T(t), \tag{A.14}$$

or,

$$\frac{d[e^{\lambda_p t} N_p(t)]}{dt} = e^{\lambda_p t} R_{T\to p} N_T(t). \tag{A.15}$$

Integrating this, after insertion of Eq. (A.11), gives

$$\begin{aligned} e^{\lambda_p t} N_p(t) &= N_0 R_{T\to p}\int dt e^{(\lambda_p - R_T)t} \\ &= \frac{N_0 R_{T\to p}}{\lambda_p - R_T} e^{(\lambda_p - R_T)t} + C, \end{aligned} \tag{A.16}$$

or,

$$N_p(t) = \frac{N_0 R_{T\to p}}{\lambda_p - R_T} e^{-R_T t} + C e^{-\lambda_p t}. \tag{A.17}$$

Inserting the initial condition $N_p(t=0) = 0$ gives

$$0 = \frac{N_0 R_{T\to p}}{\lambda_p - R_T} + C, \tag{A.18}$$

giving the solution for the parent nuclide,

$$N_p(t) = \frac{N_0 R_{T\to p}}{\lambda_p - R_T}\left[e^{-R_T t} - e^{-\lambda_p t}\right]. \tag{A.19}$$

For every nuclide i, directly produced by a nuclear reaction on the target nuclide or from decay of the parent nuclide, we have

$$\frac{dN_i(t)}{dt} = \lambda_{p\to i} N_p(t) + R_{T\to i} N_T(t) - \lambda_i N_i(t). \tag{A.20}$$

Multiplying both sides with $e^{\lambda_i t}$ gives analogous to Eqs. (A.13)-(A.15),

$$\frac{d[e^{\lambda_i t} N_i(t)]}{dt} = e^{\lambda_i t}\left[\lambda_{p\to i} N_p(t) + R_{T\to i} N_T(t)\right], \tag{A.21}$$

or, after insertion of the solutions for N_T, Eq. (A.11), and N_p, Eq. (A.19),

$$\frac{d[e^{\lambda_i t}N_i(t)]}{dt} = e^{\lambda_i t}\left[\lambda_{p\to i}\frac{N_0 R_{T\to p}}{\lambda_p - R_T}(e^{-R_T t} - e^{-\lambda_p t}) + N_0 R_{T\to i} e^{-R_T t}.\right] \tag{A.22}$$

Integrating this gives

$$\begin{aligned} e^{\lambda_i t}N_i(t) &= \frac{N_0\lambda_{p\to i}R_{T\to p}}{\lambda_p - R_T}\left[\int dt e^{(\lambda_i - R_T)t} + \int dt e^{(\lambda_i-\lambda_p)t}\right] \\ &+ N_0 R_{T\to i}\int dt e^{(\lambda_i - R_T)t} + C, \end{aligned} \tag{A.23}$$

or,

$$\begin{aligned} N_i(t) &= \frac{N_0\lambda_{p\to i}R_{T\to p}}{\lambda_p - R_T}\left[\frac{e^{-R_T t}}{\lambda_i - R_T} + \frac{e^{-\lambda_p t}}{\lambda_i - \lambda_p}\right] \\ &+ \frac{N_0 R_{T\to i}}{\lambda_i - R_T}e^{-R_T t} + Ce^{-\lambda_i t}. \end{aligned} \tag{A.24}$$

Inserting the initial condition $N_i(t=0) = 0$ gives

$$0 = \frac{N_0\lambda_{p\to i}R_{T\to p}}{\lambda_p - R_T}\left[\frac{1}{\lambda_p - R_T} + \frac{1}{\lambda_i - \lambda_p}\right] + \frac{N_0 R_{T\to i}}{\lambda_i - R_T} + C, \tag{A.25}$$

so we finally obtain

$$N_i(t) = N_0\frac{R_{T\to i}}{\lambda_i - R_T}[e^{-R_T t} - e^{-\lambda_i t}] + N_0\frac{\lambda_{p\to i}R_{T\to p}}{\lambda_p - R_T}\left[\frac{e^{-R_T t} - e^{-\lambda_i t}}{\lambda_i - R_T} - \frac{e^{-\lambda_p t} - e^{-\lambda_i t}}{\lambda_i - \lambda_p}\right]. \tag{A.26}$$

The three solutions of interest are thus given by Eqs. (A.11), (A.19), and (A.26), and these are summarized in Eq. (4.2).

A.3 Solution of the differential equations for depletion only (cooling)

After the irradiation time, T_{ir}, only decay of the various produced nuclides takes place, and the corresponding differential equations become simpler. For $t > T_{ir}$ we have for the target nuclide

$$N_T(t) = N_T(T_{ir}) \tag{A.27}$$

The parent nuclides are no longer produced from a nuclear reaction via the target (i.e. the 'R' term is no longer there) which reduces Eq. (A.9) to

$$\frac{dN_p(t)}{dt} = -\lambda_p N_p(t). \tag{A.28}$$

This is easily integrated to give

$$N_p(t) = N_p(T_{ir})\, e^{-\lambda_p(t-T_{ir})}. \tag{A.29}$$

For the nuclide of interest, Eq. (A.8) reduces to

$$\frac{dN_i(t)}{dt} = \lambda_{p\to i}N_p(t) - \lambda_i N_i(t) \tag{A.30}$$

Analogous to Eqs. (A.12) - (A.19) we can solve this to give

$$N_i(t) = \frac{N_p(T_{ir})\lambda_p}{\lambda_i - \lambda_p}\left[e^{-\lambda_p(t-T_{ir})} - e^{-\lambda_i(t-T_{ir})}\right]. \tag{A.31}$$